The starry sphere

Looking out from Earth, we see a universe full of stars. The stars are so far away that they all appear to be the same distance from us. As they move across the sky, they do so together, keeping their relative positions. The stars appear to be fixed to the inside of a giant globe enveloping Earth. Although it is a distortion of reality, this imaginary globe, called the celestial sphere, is a useful astronomical tool.

THE CELESTIAL SPHERE

The celestial sphere is important for anyone who wants to become familiar with the night sky. It is used by all observers, from beginners to experienced astronomers. It helps in understanding how location and time and date of observation determine what is in the sky. The sphere's network of lines pinpoints the positions of stars and other objects, and helps when navigating around the sky. The celestial equator, concentric with Earth's equator, divides the sphere into northern- and southern-hemisphere sky. North and south celestial poles are above Earth's poles.

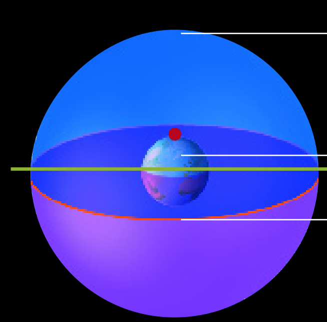
The distant stars are fixed on the sphere, and as Earth spins they appear to revolve around the poles. The much closer Sun and planets move against the backdrop of fixed stars. The Sun's path, the ecliptic, crosses the celestial equator at two points, known as the equinoxes. The Moon and planets follow paths close to the ecliptic.

YOUR VIEW OF THE CELESTIAL SPHERE

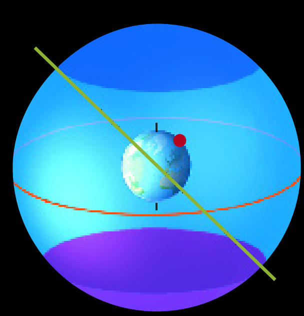
An observer's latitude determines which portion of the celestial sphere they see. Most people see all of one celestial hemisphere and part of the other, but not all of this is visible at once. As Earth makes its daily spin and yearly orbit around the Sun, new regions of the sphere come into view.

KEY TO SPHERES

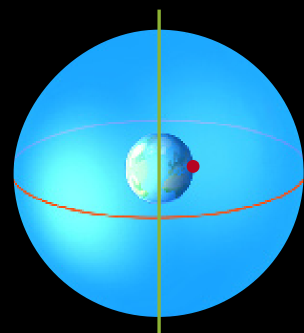
- *Horizon*
- *Observer*
- *Stars always visible*
- *Stars visible at some time*
- *Stars never visible*



View from the north pole
To an observer based at Earth's north pole, only the stars in the northern half of the celestial sphere are visible.



View from mid-latitudes
Stars close to the observer's celestial pole are always visible. Those farther from the pole are seen as Earth spins and orbits.



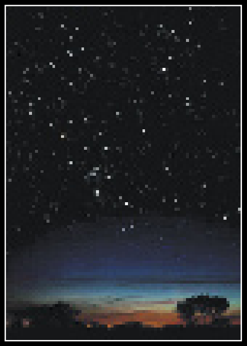
View from the equator
All parts of the celestial sphere are visible. The celestial equator is overhead, and the celestial poles are on opposite horizons.

CHANGING VIEW

From any one location on Earth it is possible to see only part of the celestial sphere at a particular time. This view changes during the evening as the Earth spins. Earth turns from west to east, and so the stars move across the sky from east to west. The view also changes over the course of a year (see pp.10–11). For instance, stars that are in the winter daytime sky can be seen in the evening sky later in the year when the Sun has moved along the ecliptic and its starry backdrop has changed.

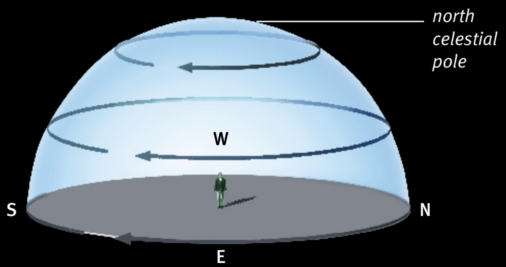
An observer's location determines not only which part of the celestial sphere is visible, but also how the stars move across the sky. Except at the equator, observers will always have part of the sky around the celestial pole above the horizon. Stars circle around the pole; the farther you are from the pole, the more circumpolar stars you will see. The position of the celestial pole in your sky corresponds to your latitude on Earth. For instance, for observers at 40°N, the north celestial pole is located 40° above their northern horizon.

ORION FROM ARIZONA

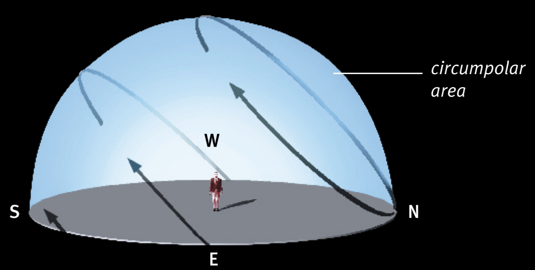


ORION FROM JAPAN

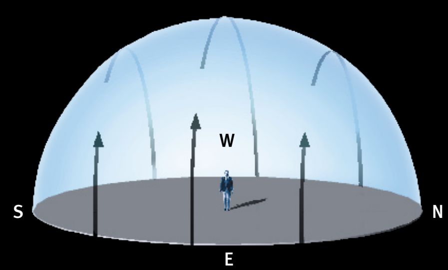
The same sky
Observers at the same latitude but on opposite sides of the world (on different longitudes) see the same sky. The constellation of Orion is clearly visible to observers in Arizona at 11 pm (left). Several hours later, once Earth has turned, the same view of Orion is seen at 11 pm in Tokyo, Japan (right).



Motion at the poles
At the north pole, the stars circle overhead, around the north celestial pole. They circle counterclockwise. At the south pole, the stars circle in the opposite direction, clockwise.



Motion at mid-latitudes
At mid-latitudes, the stars rise above the eastern horizon, cross the sky obliquely, and set below the opposite, western horizon. Also, some circumpolar stars are always in the sky.



Motion at the equator
From the equator, stars are seen to rise vertically from the eastern horizon, then move across the sky over the observer's head, before setting in the west.